Local Hessian integrability for planar Monge–Ampère inequalities

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Abstract

Let $0 < \lambda < 1$ and let u be a convex Alexandrov solution on a bounded convex planar domain, with Monge–Ampère density between λ and λ^{-1}. We prove that its full distributional Hessian belongs locally to strong L^{p_λ}, where $p_\lambda = (1+\lambda^2)/(1-\lambda^2)$. This implies the corresponding weak endpoint estimate posed in the Monge–Ampère regularity literature. The proof uses a change of variables with a Lipschitz inverse and a local weighted scalar Lipschitz projection. An energy estimate on the set where the projection differs from the original map controls the cutoff error and yields the required negative moment.

1 Introduction

The Monge–Ampère equation controls the determinant of a convex potential's Hessian. Even when that determinant is bounded above and below, its two eigenvalues can be very different. Estimates on the whole Hessian therefore contain information about anisotropy that the determinant alone does not display. This distinction is important in the regularity of convex potentials and their gradient maps; the connection with optimal transportation is described in the survey of De Philippis and Figalli [3].

For strictly convex Alexandrov solutions with two-sided density bounds, De Philippis, Figalli and Savin [4, Theorem 1.1] proved local $W^{2,1+\varepsilon}$ regularity for some positive ε depending on the density bounds and dimension. In the plane, strict convexity follows from the positive lower bound. The local Hessian is thus a matrix field, but the quantitative dependence of its integrability exponent remains a separate question. De Philippis and Figalli [3, §5.2(2)] formulate the weak endpoint exponent

$$p_\lambda = \frac{1+\lambda^2}{1-\lambda^2}, \qquad 0 < \lambda < 1,$$

under the bounds $\lambda \le \det D^2 u \le \lambda^{-1}$. We establish the following strong local estimate at this exponent.

Theorem 1.1. *Let $0 < \lambda < 1$, let $\Omega \subset \mathbb{R}^2$ be a bounded open convex domain, and let $u : \Omega \to \mathbb{R}$ be finite and convex. Suppose that its Alexandrov Monge–Ampère measure satisfies*

$$\lambda|E| \le \mu_u(E) = |\partial u(E)| \le \lambda^{-1}|E| \quad \text{for every Borel } E \subset \Omega. \tag{1.1}$$

Then the full distributional Hessian is a measurable matrix field H, and for every $\Omega' \Subset \Omega$,

$$\int_{\Omega'} (1+|H|)^{p_\lambda}\, dx < \infty, \qquad p_\lambda = \frac{1+\lambda^2}{1-\lambda^2}. \tag{1.2}$$

In particular,

$$\sup_{t>0} t^{p_\lambda} |\{x \in \Omega' : |H(x)| > t\}| < \infty. \tag{1.3}$$

Here $\partial u(x)$ is the convex subgradient, $\partial u(E)$ is the union of those subgradients over E, and $|E|$ denotes planar Lebesgue measure. The matrix norm is the operator norm. The conclusion concerns the entire distributional Hessian, including the assertion that no singular Hessian measure remains. Constants may depend on the fixed solution, λ, and the compact interior set. Chebyshev's inequality gives (1.3) from (1.2).

We use the separating function and scalar projection ideas of Agazzi et al. [1, Lemmas 3–4]. The local argument here is weighted. The periodic quasiconvexity theorem in that paper is not invoked on a local domain. The cone and reflection-map calculations follow the geometry developed by De Philippis, Guerra and Tione [5, Lemmas 6.3 and 6.9]; we give the transformations directly for the original Alexandrov class.

The change of variables $y = x + Du(x)$ turns the Hessian into the derivative of a 1-Lipschitz map. Large Hessian eigenvalues correspond to a small distance of that derivative from a circle of rotations. The inverse of each rotated difference is quasiconformal, which gives a quantitative distance deficit for the Lipschitz map on fixed interior disks. A scalar Lipschitz projection then has a uniformly small error. Its constraint is saturated wherever that error is nonzero, and the resulting energy estimate controls the distance on precisely that set. A sufficiently high power of a smooth cutoff gives an integrable error when the separating inequality is integrated over its scale.

Section 2 establishes the exact measure transformation and the derivative cone. Section 3 proves the local weighted estimate. Section 4 integrates it and returns to the Hessian. Throughout, vector norms are Euclidean and matrix norms are operator norms. We identify $\mathbb{R}^2$ with $\mathbb{C}$, and C denotes a finite local constant independent of the small projection scale and the integration cutoff.

2 Coordinates for Alexandrov solutions

We first justify the change of variables without assuming a smooth solution. The Radon–Nikodym theorem [11, Theorem 2] and (1.1) give

$$\mu_u = f\,dx, \qquad \lambda \le f \le \lambda^{-1} \quad \text{a.e.} \tag{2.1}$$

Planar strict convexity and local $C^{1,\alpha}$ regularity follow from the results recorded in [3, p. 18 and Theorem 2.14]. The normalized $W^{2,1+\varepsilon}$ theorem [4, Theorem 1.1] applies locally as follows. At x_0, subtract the supporting affine function ℓ and choose a small compact section $S = \{u - \ell < h\} \Subset \Omega$. The potential $w = u - \ell - h$ is zero on its boundary and has minimum $-h$ at x_0. John normalization [3, Lemma 2.15] gives an affine map $A(x) = Lx + c$, $\det L > 0$, with $B_1 \subset A(S) \subset B_2$. In dimension two, $(\det L)w \circ A^{-1}$ has the same density bounds: its subgradient area and its domain area both acquire the factor $\det L$. The minimum point belongs to the half-height set in the theorem of [4]. Pulling back the resulting weak derivatives gives

$$u \in C^1_{\mathrm{loc}}(\Omega) \cap W^{2,1}_{\mathrm{loc}}(\Omega).$$

Write $H = D^2u$ for this full weak matrix field. It is symmetric and positive semidefinite almost everywhere by convexity.

Proposition 2.1. *Set*

$$\psi(x) = u(x) + \tfrac{1}{2}|x|^2, \qquad Y(x) = x + Du(x), \qquad U = Y(\Omega).$$

Then $Y : \Omega \to U$ is a homeomorphism onto an open set, with 1-Lipschitz inverse F. Both maps send null sets to null sets. Moreover,

$$\det H = f, \qquad \rho = 1 + \operatorname{tr} H + \det H, \qquad |Y(E)| = \int_E \rho\,dx \tag{2.2}$$

for every Borel $E \subset \Omega$. Almost everywhere in U,

$$DF(y) = (I + H(F(y)))^{-1}. \tag{2.3}$$

Proof. Convex mollification on compact subdomains and continuity of Monge–Ampère measures [3, Proposition 2.4] give, for fixed $t \geq 0$,

$$\mu_{u+t|x|^2/2} = \mu_u + t\Delta u + t^2\, dx. \tag{2.4}$$

Indeed the smooth identity is $\det(D^2u_j + tI) = \det D^2u_j + t\Delta u_j + t^2$. The two convex measures converge weakly on compact tests, and the Laplacians converge distributionally. This proves (2.4). Consequently, with $\rho_0 = 1 + \operatorname{tr} H + f$, we have $\mu_\psi = \rho_0\, dx$.

Monotonicity of Du gives

$$(Y(x) - Y(z)) \cdot (x - z) \geq |x - z|^2.$$

Thus Y is injective and F is 1-Lipschitz. To see that U is open, take $\overline{B}_a(x_0) \subset \Omega$. On its sphere, strong convexity gives

$$\psi(x) - y \cdot x - \psi(x_0) + y \cdot x_0 \geq a^2/2 - |y - Y(x_0)|a.$$

For $|y - Y(x_0)| < a/2$, the minimum on this ball is interior, so $D\psi = y$ somewhere in the ball. This also proves continuity of the inverse on the image. Since $\psi \in C^1$, its subgradient is Y, and

$$|Y(E)| = \int_E \rho_0\, dx. \tag{2.5}$$

Hence Y sends null sets to null sets. The Lipschitz map F does too, by the area formula [7, §2, Theorem 1].

The convex Legendre supremum $\psi^*(y) = \sup_{x\in\Omega}(x \cdot y - \psi(x))$ is finite: Ω is bounded and ψ has an affine lower bound. For $y \in U$, its unique maximizer is $F(y)$. The inequalities

$$F(y) \cdot h \leq \psi^*(y + h) - \psi^*(y) \leq F(y + h) \cdot h$$

show that $D\psi^* = F$ on U, since the remainder is at most $|h|^2$. Thus $G = DF$, defined almost everywhere by Rademacher's theorem [10, Theorem 2.2.4], is symmetric and positive semidefinite. Let B consist of its singular-derivative set and the null set where the derivative does not exist. The area formula gives $|F(B)| = 0$; (2.5) then gives $|B| = |Y(F(B))| = 0$. Therefore G is nonsingular almost everywhere.

At such a point y, the expansion $F(y + h) - F(y) = Gh + o(|h|)$ implies a lower bound $c|h|$. Continuity of the inverse converts this expansion into differentiability of Y at $F(y)$, with derivative G^{-1}. The exceptional set has null image under F. The ordinary derivative of this continuous $W^{1,1}_{\mathrm{loc}}$ map agrees almost everywhere with its weak derivative $I + H$. One can verify this on almost every coordinate line: Fubini gives the one-dimensional weak derivative, its integral is an absolutely continuous representative, continuity identifies that representative, and Lebesgue differentiation identifies its ordinary derivative. These integration and differentiation statements are given in [10, Theorems 1.6.11–12 and 1.7.21]. The null-image property of Y now gives (2.3) in image coordinates.

Applying the area formula to F on $Y(E)$, and using the weighted form of (2.5), yields

$$|E| = \int_{Y(E)} \det G\, dy = \int_E \frac{\rho_0}{\det(I + H)}\, dx.$$

Equality on every Borel set implies $\rho_0 = \det(I + H) = 1 + \operatorname{tr} H + \det H$, so $\det H = f$. This proves (2.2). To apply the globally stated area formula, each component of F may be extended to $\mathbb{R}^2$ by the infimum Lipschitz extension. Its general multiplicity formula is then restricted to sets in U, where F is injective. The extension need not be injective outside U. □

Define

$$W(y) = y - 2F(y), \qquad v(y) = A_0 W(y), \qquad A_0 = \mathrm{diag}(1, -1). \tag{2.6}$$

For $x, z \in \Omega$, monotonicity gives

$$|W(Y(x)) - W(Y(z))|^2 = |Y(x) - Y(z)|^2 - 4(Du(x) - Du(z)) \cdot (x - z) \le |Y(x) - Y(z)|^2.$$

Thus W and v are 1-Lipschitz. Their derivatives satisfy

$$DW(Y(x)) = (H - I)(H + I)^{-1}. \tag{2.7}$$

For a real matrix M, write its complex action as $Mz = a_+(M)z + a_-(M)\bar{z}$. Its operator norm and determinant are

$$|M| = |a_+(M)| + |a_-(M)|, \qquad \det M = |a_+(M)|^2 - |a_-(M)|^2.$$

Since DW is symmetric, the derivative at almost every $y \in U$ acts by

$$Dv(y)\xi = b(y)\xi + a(y)\bar{\xi} \quad (\xi \in \mathbb{C}), \qquad b = b_1 - ib_2, \quad a \in \mathbb{R}. \tag{2.8}$$

We suppress the base point in the coefficient functions below. Put

$$K = \lambda^{-1}, \qquad k = \frac{K - 1}{K + 1}, \qquad d = 1 - |b|. \tag{2.9}$$

Proposition 2.2. *Almost everywhere in U,*

$$0 < d \le 1, \qquad |a| \le kd, \qquad |Dv| \le 1 - (1 - k)d. \tag{2.10}$$

If $h_1 \ge h_2 > 0$ are the eigenvalues of H, then

$$\frac{2}{(1 + k)(1 + h_1)} \le d(Y(x)) \le \frac{2K}{1 + h_1}, \qquad 1 + h_1 \le \rho \le C_K(1 + h_1). \tag{2.11}$$

For every rotation $Q \in \mathrm{SO}(2)$, the map $g_Q(y) = v(y) - Qy$ is a K-quasiconformal homeomorphism onto its image.

Proof. The eigenvalues of DW are $(h_i - 1)/(h_i + 1) \in (-1, 1)$; therefore $|b| + |a| < 1$. The inverse Cayley formula gives

$$\det H = \frac{(1 + a)^2 - |b|^2}{(1 - a)^2 - |b|^2}.$$

The determinant band is equivalent to $2|a| \le k(1 + a^2 - |b|^2)$. Hence

$$|a| \le \frac{k(1 - |b|)(1 + |b|)}{2 - k|a|} \le k(1 - |b|),$$

where the last inequality follows from $|b| + k|a| \le 1$. This proves (2.10). The matrix computation is the operator-norm cone calculation in [5, Lemma 6.3].

The operator distance of DW from symmetric orthogonal reflections is

$$d + |a| = \max\left\{\frac{2}{1 + h_1}, \frac{2h_2}{1 + h_2}\right\}.$$

To check the formula, a reflection has eigenvalues $1, -1$. Its two eigendirections give lower bounds $1 - \alpha_1$ and $1 + \alpha_2$, where $\alpha_1 \ge \alpha_2$ are the eigenvalues of DW; aligning these directions with those

of DW attains their maximum. Equivalently, left multiplication by A_0 turns this distance into $\min_{|Q|=1}(|b-Q|+|a|) = d+|a|$. Since $h_2 = f/h_1$,

$$\frac{h_2(1+h_1)}{1+h_2} = \frac{f(1+h_1)}{h_1+f} \le \max\{f,1\} \le K.$$

Together with $d \le d+|a| \le (1+k)d$, this proves the distance comparison in (2.11). The density comparison follows from $\rho = 1+h_1+h_2+f$, $h_2 \le h_1$, and $f \le K$.

The complex coefficients of Dg_Q are $b-Q, a$, and $|a| \le kd \le k|b-Q|$. Thus $|Dg_Q|^2 \le K \det Dg_Q$, with positive determinant. To establish homeomorphism, set $S = A_0 Q$, a symmetric reflection. Choose orthonormal $e, e^\perp$ with $Se = e$, $Se^\perp = -e^\perp$. In original coordinates,

$$W(Y(x)) - SY(x) = -2(x\cdot e)e + 2(Du(x)\cdot e^\perp)e^\perp.$$

Equal outputs fix $x\cdot e$. On that line, $Du\cdot e^\perp$ is strictly increasing by differentiable strict convexity, so the inputs coincide. The map is also open: in a small rectangle about any point, the derivative values at the upper and lower ends of its central vertical segment strictly surround the central value. Continuity preserves a smaller such interval for all nearby horizontal coordinates, and the intermediate value theorem fills an open output rectangle. This proves homeomorphism after composition with F and A_0. It is the C^1 fiber argument underlying [5, Lemma 6.9]. The analytic quasiconformal convention [5, §2.1] is precisely $|Dg_Q|^2 \le K \det Dg_Q$ for a $W^{1,2}_{\mathrm{loc}}$ homeomorphism, so its K agrees with (2.9). □

3 A local weighted projection estimate

3.1 A distance deficit on interior disks

Fix concentric disks $\overline{B}_r(y_0) \subset B_R(y_0) \Subset U$, with $0 < r < R$. All constants in this section may depend on these disks and the fixed map.

Lemma 3.1. *There is $c > 0$ such that, for all $y, z \in \overline{B}_r(y_0)$,*

$$|v(y) - v(z)| \le |y-z| - c|y-z|^K. \tag{3.1}$$

Proof. Joint continuity, compactness of $\mathrm{SO}(2)$, and the injectivity from Proposition 2.2 give

$$\ell = \min_{\substack{Q\in\mathrm{SO}(2),\, y\in\overline{B}_r \\ z\in\partial B_R}} |g_Q(y) - g_Q(z)| > 0.$$

The image $V_Q = g_Q(B_R)$ is bounded and simply connected, and $\partial V_Q = g_Q(\partial B_R)$. These facts follow by pulling loops back into the disk and using the homeomorphism on its compact closure. Choose a Riemann map $\chi_Q : V_Q \to \mathbb{D}$ with $\chi_Q(g_Q(y_0)) = 0$, using [8, p. 1]. The disk self-homeomorphism $T_Q(z) = \chi_Q(g_Q(y_0 + Rz))$ and its inverse are K-quasiconformal and fix zero. Inversion and conformal composition preserve K; see [2, Theorem 4.16 and the following paragraph]. Mori's theorem [6, Theorem 1.1], in its planar normalized form, states

$$|T_Q^{-1}(\xi) - T_Q^{-1}(\xi')| \le 16|\xi - \xi'|^{1/K}.$$

The set $g_Q(\overline{B}_r)$ is at boundary distance at least ℓ in V_Q. If two such image points have distance $\sigma < \ell/2$, their connecting segment is inside V_Q, at distance at least $\ell/2$ from its boundary. Cauchy's

derivative formula [9, p. 5], on circles of radius $\ell/4$, gives $|\chi_Q'| \le 4/\ell$ along this segment. Mori therefore gives

$$|y-z| \le 16R(4\sigma/\ell)^{1/K}.$$

If $\sigma \ge \ell/2$, use $|y-z| \le 2r$. Thus, uniformly in Q,

$$|y-z| \le C|g_Q(y) - g_Q(z)|^{1/K}. \tag{3.2}$$

For a pair with $v(y) \ne v(z)$, choose Q taking the direction of $y-z$ to that of $v(y)-v(z)$. Because v is 1-Lipschitz, its image difference under g_Q then has length $|y-z| - |v(y)-v(z)|$. Raising (3.2) to power K proves (3.1). When the vector difference vanishes, decrease c to at most $(2r)^{1-K}$; the same inequality holds. □

3.2 The scalar projection

Let $f_1 = v_1$. Choose $\zeta \in C_c^\infty(B_r)$, $0 \le \zeta \le 1$, equal to one on $B_{r/2}$. Fix an integer $m > K+1$ and set $\eta = \zeta^m$. The high power will be used to control $D\eta$. For $0 < \epsilon < 1/2$, put $s = 1-\epsilon$. Choose a scalar s-Lipschitz minimizer w of

$$\int_{B_r} \eta(f_1 - w)^2,$$

and set $h = f_1 - w$. The construction adapts the projection in [1, Lemma 4] to this local weight.

Lemma 3.2. *The minimizer can be chosen so that*

$$\|h\|_\infty \le C\epsilon^{K/(K-1)}. \tag{3.3}$$

On $E = \{h \ne 0\} \cap \{\eta > 0\}$, $|Dw| = s$ almost everywhere. On $\{h = 0\}$, $Dh = 0$ almost everywhere. In addition,

$$\int \eta Dw \cdot Dh \ge -\int hD\eta \cdot Dw. \tag{3.4}$$

Proof. Existence follows from compactness of bounded equi-Lipschitz scalar functions on the closed disk. The positive mass of η and a bounded objective bound one value of a minimizing sequence, and the Lipschitz constraint bounds all its values. A uniformly convergent subsequence has an admissible limit with the limiting objective. The variational inequality is

$$\int \eta h(\xi - w) \le 0 \quad \text{for every scalar } s\text{-Lipschitz competitor } \xi. \tag{3.5}$$

Define the two s-Lipschitz envelopes

$$f_s^-(y) = \inf_{z \in \overline{B}_r} (f_1(z) + s|y-z|), \qquad f_s^+(y) = \sup_{z \in \overline{B}_r} (f_1(z) - s|y-z|).$$

They satisfy $f_s^- \le f_1 \le f_s^+$. By Lemma 3.1, either deviation from f_1 is bounded by $\sup_{t \ge 0}(\epsilon t - ct^K) \le C\epsilon^{K/(K-1)}$. Clipping a minimizer between these envelopes preserves its Lipschitz constraint and does not increase its squared distance from f_1. This proves (3.3).

At a differentiability point where $h > 0$, $\eta > 0$, and $|Dw| < s$, a small cone $w(y_1) + t - s|y - y_1|$ lies below w on a sufficiently small surrounding sphere and above it at the center. Replace w by its maximum with that cone inside the sphere. The functions agree at the boundary; splitting line segments there shows that the modification remains s-Lipschitz on the original disk. It increases w on an open subset where $\eta h > 0$, contradicting (3.5). The minimum with an upward cone handles

$h < 0$. This proves saturation almost everywhere. The zero-set assertion follows at its density and differentiability points: a nonzero derivative would produce a conical sector of nonzero values, contradicting density one.

Extend w to $\mathbb{R}^2$ with the same Lipschitz bound. Both translations $w(\cdot + te_i)$ and $w(\cdot - te_i)$, restricted to the disk, are competitors in (3.5). With $A = \eta h$, extended by zero, their summed inequality is

$$\begin{aligned} 0 &\geq \int A(y)\big(w(y+te_i) + w(y-te_i) - 2w(y)\big)\,dy \\ &= -\int \big(A(y+te_i) - A(y)\big)\big(w(y+te_i) - w(y)\big)\,dy. \end{aligned}$$

Divide by t^2. The first differences converge almost everywhere to the respective first derivatives, and are bounded on a fixed compact set. Dominated convergence gives $\int \partial_i A\, \partial_i w \geq 0$. Summing over i proves (3.4). Only first differences were used; compact support controls the translations without a boundary term. □

3.3 The separating inequality and energy estimate

For a real matrix M, define

$$L_s(M) = \begin{cases} \det M, & |M| \leq s, \\ 2s|a_+(M)| - s^2, & |M| > s, \end{cases} \qquad D_\epsilon(M) = L_{1-\epsilon}(M) - \det M. \tag{3.6}$$

Here $a_+(M)$ is the coefficient of z in $Mz = a_+(M)z + a_-(M)\bar{z}$. This is the rescaled separating function of [1, Lemma 3]. We record the pointwise calculation that will be used.

Lemma 3.3. *If M_1, M_2 are the rows of M, then*

$$L_s(M) \geq \det(z, M_2) + z \cdot (M_1 - z)$$

whenever $|z| = s$, or whenever $z = M_1$ and $|M_1| \leq s$.

Proof. Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. For $|z| = s$, the supremum of the right side is $s|M_1 - JM_2| - s^2 = 2s|a_+(M)| - s^2$. Write $\alpha = |a_+(M)|$, $\tau = |a_-(M)|$. When $|M| \leq s$, the difference between $\det M$ and this supremum is $(s-\alpha)^2 - \tau^2 \geq 0$, because $s \geq \alpha + \tau$. When $|M| > s$, the definition already equals the supremum. For $z = M_1$, the right side is $\det M$. If $|M| > s$ but $|M_1| \leq s$, then $|\alpha - \tau| \leq |M_1| \leq s < \alpha + \tau$; hence $(s-\alpha)^2 - \tau^2 \leq 0$, so $L_s(M) \geq \det M$ also in this case. □

Proposition 3.1. *For the fixed disks and cutoff above, let*

$$\gamma = \frac{1}{K-1} - \frac{1}{m}, \qquad \beta = \frac{2}{K^2 - 1}.$$

Then $\gamma > \beta$, and for all $0 < \epsilon < 1/2$,

$$\int \eta D_\epsilon(Dv)\,dy \geq -C\epsilon^{2+\gamma}. \tag{3.7}$$

Proof. By Lemma 3.2, Dw has length s on E, and equals Df_1 where $h = 0$. Lemma 3.3 therefore gives

$$D_\epsilon(Dv) \geq -\det(Dh, Dv_2) + Dw \cdot Dh \quad \text{on } \{\eta > 0\}.$$

The real-antilinear form (2.8) gives the rows

$$Df_1 = (b_1 + a, b_2), \qquad Dv_2 = (-b_2, b_1 - a).$$

Thus $B = -JDv_2 = Df_1 - 2ae_1$ is distributionally divergence free. Testing against ηh and using (3.4),

$$\begin{aligned}\int \eta D_\epsilon(Dv) &\geq \int hD\eta \cdot (Df_1 - Dw) - 2\int ha\,\partial_1\eta \\ &= -\frac{1}{2}\int h^2\Delta\eta - 2\int ha\,\partial_1\eta.\end{aligned} \tag{3.8}$$

These integrations by parts use compactly supported Lipschitz tests and bounded first derivatives. Smooth approximation of the tests justifies them in the weak sense.

To estimate the last term, use saturation on its actual nonzero set. The identity $Df_1 = Dw + Dh$ gives

$$2\int \eta Dw \cdot Dh + \int \eta|Dh|^2 = \int_E \eta(|Df_1|^2 - s^2).$$

From (2.10), $|Df_1| \leq 1 - (1-k)d$, while $s^2 \geq 1 - 2\epsilon$. Since $(1-t)^2 \leq 1 - t$ for $0 \leq t \leq 1$,

$$|Df_1|^2 - s^2 \leq 2\epsilon - (1-k)d.$$

Combining this with (3.4) and (3.3) yields

$$(1-k)\int_E \eta d + \int \eta|Dh|^2 \leq 2\epsilon\int_E \eta + 2\|h\|_\infty \int |D\eta| \leq C\epsilon. \tag{3.9}$$

The chosen power of the cutoff satisfies $|D\eta| \leq C\eta^{1-1/m}$. Hölder's inequality and $d \leq 1$ now give

$$\int_E d|D\eta| \leq C\left(\int_E \eta d\right)^{1-1/m}\left(\int_E d\right)^{1/m} \leq C\epsilon^{1-1/m}.$$

Since $|a| \leq kd$ and $\|h\|_\infty \leq C\epsilon^{1+1/(K-1)}$, the two terms in (3.8) satisfy

$$\left|\int ha\,\partial_1\eta\right| \leq C\epsilon^{2+1/(K-1)-1/m}, \qquad \left|\int h^2\Delta\eta\right| \leq C\epsilon^{2+2/(K-1)}.$$

This proves (3.7). Finally,

$$\gamma - \beta = \frac{1}{K+1} - \frac{1}{m} > 0,$$

by the choice $m > K+1$. □

4 Negative moments and Hessian integrability

We integrate (3.7) over its scale. The sign of that integral must be controlled uniformly for every positive lower cutoff.

Lemma 4.1. *Let $0 < d \leq 1$, $0 \leq A \leq kd$, and define*

$$P_\epsilon = \begin{cases} 0, & \epsilon \leq d - A, \\ A^2 - (d-\epsilon)^2, & \epsilon > d - A. \end{cases}$$

For every $0 < q < (1-k)/k$, there are constants $c_q > 0, C_q < \infty$, depending only on q, k, such that

$$\int_{h_0}^{1/2} P_\epsilon\,\epsilon^{-q-3}\,d\epsilon \leq C_q - c_q \max\{d, h_0\}^{-q}, \qquad 0 < h_0 < 1/2. \tag{4.1}$$

Proof. Put $b_0 = A/d \in [0,k]$. Direct integration of the three monomials gives

$$\begin{aligned} I_0 &= \int_{d-A}^{\infty} [A^2 - (d-\epsilon)^2]\epsilon^{-q-3}\,d\epsilon \\ &= \frac{2((q+1)b_0 - 1)}{q(q+1)(q+2)}(1-b_0)^{-q-1}d^{-q} \le -c_0 d^{-q}, \end{aligned}$$

where $c_0 = 2(1-(q+1)k)/(q(q+1)(q+2)) > 0$. All denominators are positive, since $d - A \ge (1-k)d > 0$. If $h_0 \le d - A$, the continued integral from h_0 equals I_0. If $d - A < h_0 \le d + A$, the removed part is nonnegative, so the remainder is at most I_0. In either case it is bounded above by $-c_0 \max\{d, h_0\}^{-q}$.

If $h_0 > d + A$, the remaining integrand is negative. For $\epsilon \ge 4h_0$, using $A, d \le h_0$, we have

$$A^2 - (d-\epsilon)^2 \le -\epsilon^2/2.$$

Its tail is at most $-(4h_0)^{-q}/(2q)$. Finally, the absolute integral of the continued expression from $1/2$ to infinity is bounded uniformly in $d, A \le 1$, because $|A^2 - (d-\epsilon)^2| \le C(1+\epsilon^2)$ and $q > 0$. Removing that tail proves (4.1). This continuation is only a scalar polynomial calculation. □

Proof of Theorem 1.1. For the map v from Section 2, the definition (3.6) gives, with $A = |a|$,

$$D_\epsilon(Dv) = 0 \quad (\epsilon \le d - A), \qquad D_\epsilon(Dv) = A^2 - (d-\epsilon)^2 \quad (\epsilon > d - A).$$

Take $q = \beta = 2/(K^2 - 1)$. Then

$$0 < q < \frac{1-k}{k} = \frac{2}{K-1}, \qquad \gamma > q.$$

By Proposition 3.1,

$$\int_{h_0}^{1/2} \left(\int \eta D_\epsilon(Dv)\,dy \right) \epsilon^{-q-3}\,d\epsilon \ge -C \int_0^{1/2} \epsilon^{\gamma - q - 1}\,d\epsilon > -\infty,$$

uniformly in $h_0 > 0$. At each positive cutoff, the integrand is absolutely integrable on the product of $\operatorname{supp}\eta$ and $[h_0, 1/2]$. Fubini's theorem, in its absolute-integrability form [10, Theorem 1.7.21 and Corollary 1.7.23], and Lemma 4.1 therefore imply a uniform bound on $\int \eta \max\{d, h_0\}^{-\beta}\,dy$. As $h_0 \downarrow 0$, these nonnegative functions increase. Monotone convergence [10, Theorem 1.4.44] gives

$$\int_{B_{r/2}} d^{-\beta}\,dy < \infty. \tag{4.2}$$

No measurable choice of the projections as ϵ varies is required: the integrated expression is the explicit $D_\epsilon(Dv)$.

For $\Omega' \Subset \Omega$, the set $Y(\overline{\Omega'})$ is compact in U. Finitely many inner disks of the kind above cover it, so $\int_{Y(\Omega')} d^{-\beta}\,dy < \infty$. The weighted inverse area formula [7, §2, Theorem 2], Proposition 2.1, and (2.11) give

$$\begin{aligned} \int_{\Omega'} (1 + |H|)^{1+\beta}\,dx &= \int_{Y(\Omega')} \frac{(1 + h_1(F(y)))^{1+\beta}}{\rho(F(y))}\,dy \\ &\le C_K \int_{Y(\Omega')} d(y)^{-\beta}\,dy < \infty. \end{aligned}$$

Indeed $\rho \ge 1 + h_1$ and $d \le 2K/(1+h_1)$. Since $1 + \beta = (1+\lambda^2)/(1-\lambda^2)$, this is (1.2). Finally, $t^{p_\lambda} \mathbf{1}_{\{|H| > t\}} \le |H|^{p_\lambda}$ for every $t > 0$, proving (1.3). □

AI-use statement

OpenAI Codex was used with substantial involvement in developing the argument, checking source applications, and preparing the exposition. An AI reader and an AI referee, each in a fresh context, reviewed the same initial anonymous manuscript. Their reports concern that snapshot; one author revision followed with AI assistance. Independent mathematical review is requested.

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